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GROUP REPORT

Operational energy Efficiency for Users (OEU); Constraints and best practices for wireless network infrastructure in educational environments

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Contents

Intellectual Property Rights 5
Foreword 5
Modal verbs terminology 5
Executive summary 5
Introduction 6
1 Scope 7
2 References 7
2.1 Normative references 7
2.2 Informative references 7
3 Definition of terms, symbols and abbreviations 8
3.1 Terms 8
3.2 Symbols 8
3.3 Abbreviations 8
4 Alternative wireless technologies to be used in schools 9
4.1 Presentation 9
4.2 Unwired buildings 9
4.2.1 PLC embedding Wi-Fi® 9
4.2.1.1 Fundamental Principles 9
4.2.1.2 Standardization 9
4.2.1.3 Constraints Related to Buildings' Topology and Uses 10
4.2.1.4 General Recommendations 10
4.2.1.5 Specific needs of French Elementary Schools 10
4.2.2 PLC + Li-Fi 10
4.2.2.1 Li-Fi Fundamental Principles 10
4.2.2.2 Standardization 11
4.2.2.3 Constraints Related to Buildings' Topology and Uses 11
4.2.2.4 General Recommendations 11
4.2.2.5 Specific Needs of French Elementary Schools 11
4.3 Wired buildings 11
4.3.1 RJ45 cabling + Wi-Fi® Hotspot 11
4.3.1.1 Fundamental Principles 11
4.3.1.2 Standardization 11
4.3.1.3 Constraints Related to Buildings' Topology and Uses 12
4.3.1.4 General Recommendations 12
4.3.1.5 Specific Needs of French Elementary Schools 12
4.3.2 RJ45 cabling + Li-Fi 12
4.3.2.1 Fundamental Principles 12
4.3.2.2 Specific Needs of French Elementary Schools 12
5 Specific configurations to be used in French Schools 13
5.1 Official constraints 13
5.1.1 Abeille Law requirements (extract) 13
5.1.2 Ministry of Education 13
5.2 METHOD #1 – Micro Smart Plug 13
5.2.1 Fundamental Principles 13
5.2.2 General Recommendations 13
5.3 Method #2 - Centralized Wi-Fi® management using a controller 14
5.3.1 Wi-Fi Hotspot managed with Control command 14
5.3.1.1 Fundamental Principles 14
5.3.1.2 Standardization 14
5.3.1.3 Constraints Related to Buildings' Topology and Uses 14
5.3.1.4 General Recommendations 14
5.3.1.5 Specific Needs of French Elementary Schools 14
5.4 Method #3 – Decentralized Wi-Fi® management using unconnected buttons 14

5.4.1 Decentralized Wi-Fi® management 14
5.4.1.1 Fundamental Principles 14
5.4.1.2 Standardization 14
5.4.1.3 Constraints Related to Buildings' Topology and Uses 15
5.4.1.4 General Recommendations 15
6 Conclusion 15
Annex A: Guidelines for the use of Wi-Fi® in establishments and schools - Technical framework 16
Annex B: Guidelines for the use of Wi-Fi® in establishments and schools - Radio frequencies and health 17
Annex C: Guidelines for the use of Wi-Fi® in establishments and schools - Uses and legal framework 18
History 20

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Foreword

This Group Report (GR) has been produced by ETSI Industry Specification Group (ISG) Operational energy Efficiency for Users (OEU).

Modal verbs terminology

In the present document "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the ETSI Drafting Rules (Verbal forms for the expression of provisions).

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Executive summary

An increasing number of people are experiencing hypersensitivity to electromagnetic waves. Young learners are more sensitive to radio waves because of their morphology, particularly the thinness of their skull. Although no research has proven the dangers of electromagnetic waves, adhering to a precautionary principle is essential, as reflected in France by the "Abeille Law" [i.13]".

Local authorities and municipalities should prioritize energy savings, particularly in schools. Therefore, Wi-Fi® should be turned off when not in use to reduce electricity consumption, just as lights should be turned off when not needed.

Minimizing exposure to radio waves and improving energy efficiency are two key challenges in developing eco-friendly, health-conscious wireless networks. The present document reviews alternative wireless solutions that address these concerns, with a particular focus on Wi-Fi® infrastructure, which is currently more widely deployed.

A point is devoted to Li-Fi, highlighting its key advantages and disadvantages for use in schools, given the current state of the technology.

Unwired buildings:

- PLC embedding Wi-Fi
- PLC + Li-Fi

Wired buildings:

- RJ45 cabling + Wi-Fi Hotspot
- RJ45 cabling + Li-Fi

The final point of the present document outlines the requirements of the Abeille Law [i.13] for implementing compliant Wi-Fi networks in French elementary schools. A key rule is that the Wi-Fi, called "Disengageable Wi-Fi", should be turned off once educational activities end. To achieve this, three technical methods are proposed to effectively disable Wi-Fi Hotspots while minimizing both wiring and battery use:

- **Method#1 - **Controlled Plug Method****: Using Controlled Plug is briefly described to explain why it is not recommended.
- **Method #2 - **Centralized Wi-Fi Management****: Using the IP protocol, each classroom has a physical or virtual controller switch connected to the Local Area Network (LAN). A centralized Wi-Fi control device receives commands from these controller switches and responds by sending IP control commands to each Wi-Fi Hotspot serving its respective classroom, instructing each hotspot to turn off its Wi-Fi emissions.
- **Method #3 - **Decentralized Wi-Fi Management****: Each classroom has a physical or virtual control switch that uses Smart Building protocols (such as Zigbee® and EnOcean® [i.1]) to communicate locally with devices, enabling Wi-Fi Hotspots in the classroom to be turned on or off.

Introduction

Each alternative technology or infrastructure is presented to help end-users choose the most appropriate one:

- Fundamental Principles
- Standardization
- Constraints Related to Buildings' Topology and Uses
- General Recommendations
- Specific Needs of French Elementary Schools.

1 Scope

The present document aims to discuss in greater detail the solutions that meet the needs of French elementary schools and, more generally, those of organizations seeking to reduce exposure to radio waves and limit energy consumption. The present document does not intend to detail every technology and deployment methodology.

2 References

2.1 Normative references

Normative references are not applicable in the present document.

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, ETSI cannot guarantee their long-term validity.

The following referenced documents may be useful in implementing an ETSI deliverable or add to the reader's understanding, but are not required for conformance to the present document.

[i.1] Enocean® alliance: "EnOcean Wireless Standard".

[i.2] IEEE 802.11bb™: "IEEE Standard for Information Technology--Telecommunications and Information Exchange between Systems Local and Metropolitan Area Networks--Specific Requirements Part 11: Wireless".

[i.3] Recommendation ITU-T G.9991: "High-speed indoor visible light communication transceiver - System architecture, physical layer and data link layer specification".

[i.4] ISO/IEC 14543-3-10: "Information technology — Home electronic systems (HES) architecture — Part 3-10: Amplitude modulated wireless short-packet (AMWSP) protocol optimized for energy harvesting — Architecture and lower layer protocols".

[i.5] ITU-T Technical Paper (09/2020): "GSTP-HNSG Technical paper on the use of G.hn technology for smart grid".

[i.6] IEEE: "HomePlug® AV2" Version 2.1 February 21, 2014.

[i.7] IEEE Std 802.3af™-2003: "IEEEE Standard for Information Technology - Telecommunications and Information Exchange Between Systems - Local and Metropolitan Area Networks - Specific Requirements - Part 3: Carrier Sense Multiple Access with Collision Detection (CSMA/CD) Access Method and Physical Layer Specifications - Data Terminal Equipment (DTE) Power Via Media Dependent Interface (MDI)".

[i.8] IEEE Std 802.3at™-2009: "IEEE Standard for Information technology-- Local and metropolitan area networks -- Specific requirements -- Part 3: CSMA/CD Access Method and Physical Layer Specifications Amendment 3: Data Terminal Equipment (DTE) Power via the Media Dependent Interface (MDI) Enhancements".

[i.9] IEEE 802.3bt™-2018: "IEEE Standard for Ethernet Amendment 2: Physical Layer and Management Parameters for Power over Ethernet over 4 pairs".

[i.10] "L'école change avec le numérique / Référentiel sur l'usage du Wi-Fi en établissement et école - Cadre technique", Version 1.0, Mai 2015.

[i.11] "L'école change avec le numérique / Référentiel sur l'usage du Wi-Fi en établissement et école - Radiofréquences et santé", Version 1.0, Mai 2015.

[i.12] "L'école change avec le numérique / Référentiel sur l'usage du Wi-Fi en établissement et école - Usages et cadre juridique", Version 1.0, Mai 2015.

[i.13] LOI n° 2015-136 du 9 février 2015 relative à la sobriété, à la transparence, à l'information et à la concertation en matière d'exposition aux ondes électromagnétiques (Abeille Law).

[i.14] Ministère de l'Education National: "Référentiel Wi-Fi".

[i.15] NIST FIPS 197: "Advanced Encryption Standard (AES)".

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the following terms apply:

radon: radioactive noble gas is important to be measured inside buildings

3.2 Symbols

Void.

3.3 Abbreviations

For the purposes of the present document, the following abbreviations apply:

AC/DC	Alternating Current / Direct Current
AES	Advanced Encryption Standard

NOTE: Standard cryptographic algorithm specified in FIPS-197 [i.15].

ANSES	Agence nationale de sécurité sanitaire de l'alimentation, de l'environnement et du travail. The French Agency for Food, Environmental and Occupational Health & Safety
BYOD	Bring You Own Device
CCMP	Counter mode with Cipher block chaining Message authentication code Protocol
CO_2	Carbon dioxide
EAP	Extensible Authentication Procedure
ETSI	European Telecommunications Standards Institute
IEC	International Electrotechnical Commission
IEEE	Institute of Electrical and Electronics Engineers
IoT	Internet of Things
IP	Internet Protocol
ISO	International Organization for Standardization
IT	Information Technology
ITU	International Telecommunication Union
LAN	Local Area Network
LED	Light-Emitting Diode
Li-Fi	Light Fidelity
N	Neutral

NOTE: Return path for electric current.

OEU	Operational energy Efficiency for Users
PC	Personal Computer
PE	Protective Earth
PLC	Power Line Communication
PoE	Power over Ethernet

RJ45	Registered Jack 45
SAR	Specific Absorption Rates
USB	Universal Serial Bus
WEP	Wireless Encryption Protocol
Wi-Fi®	Wireless Fidelity
WPA	Wi-Fi® Protected Access
WPA2	Wi-Fi® Protected Access 2

4 Alternative wireless technologies to be used in schools

4.1 Presentation

The demand for mobility requires high-speed access in schools. Older buildings often lack adequate RJ45 wiring. By using existing electrical wiring, PLC technology offers an environmentally friendly and cost-effective solution. Both Wi-Fi® and Li-Fi could benefit from this.

4.2 Unwired buildings

4.2.1 PLC embedding Wi-Fi®

4.2.1.1 Fundamental Principles

A PLC injector is installed inside the main electrical cabinet, above the three low-voltage electrical lines, the Neutral (N), and the Protective Earth (PE) of the General Power Outlet Circuit Breaker.

On the user's side, tablets and PC are wirelessly connected to the LAN through PLC/Wi-Fi plugs.

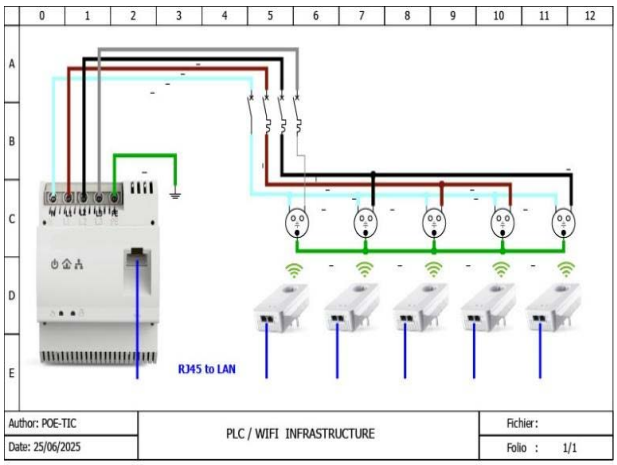


Figure 1: PLC principle

4.2.1.2 Standardization

The main standards for Power Line Communication (PLC) are Homeplug AV2 [i.6] and G.hn [i.5]. Both standards are not compatible but can coexist in the same building.

G.hn [i.5] is currently the most popular option.

4.2.1.3 Constraints Related to Buildings' Topology and Uses

Depending on the building surfaces and the number of electrical cabinets, multiple PLC injectors may be required. In such cases, RJ45 wiring should be installed in each electrical cabinet that requires a PLC injector.

The number of users' PCs and tablets connecting simultaneously is crucial. The PLC infrastructure's bandwidth should be sufficient.

Each PLC/Wi-Fi plug serves as a node, and it is essential to limit the number of nodes, especially when using consumer products.

It is important to note that PLC bandwidth is shared across the entire electrical installation. Therefore, increasing the number of injectors may significantly reduce overall performance.

4.2.1.4 General Recommendations

The use of PLC technology should be limited to small buildings, such as elementary schools and kindergartens.

Even, if possible, it is necessary to avoid having multiple PLC protocols coexisting in the same electrical installation, as this could limit the installation's bandwidth.

4.2.1.5 Specific needs of French Elementary Schools

In France, a specific law requires Wi-Fi to be turned off at the end of digital educational activities (see details in clause 5). Each PLC/Wi-Fi plug has a button to turn off Wi-Fi signals, ensuring compliance with legal requirements. However, this requires that PLC devices are easily accessible to users. This is not the case when PLC plugs are installed at the bottom of the wall. Additionally, it is not recommended to use the electrical outlet at middle height because it is at the same level as children's heads.

Furthermore, PLC provides limited bandwidth, which may be insufficient when a class of 30 students needs to connect their tablets at the same time.

4.2.2 PLC + Li-Fi

4.2.2.1 Li-Fi Fundamental Principles

The fundamental principle of Li-Fi is using optical signals to transfer data within a network. Li-Fi devices rely on Light-Emitting Diodes (LEDs) to transmit data by modulating the light's frequency. These Li-Fi transmitters, installed in the ceiling and connected to the Local Area Network (LAN) via a Li-Fi Access Point, communicate with receivers linked to computers, tablets, or Internet of Things (IoT) devices.

Light cannot pass through walls, ensuring high security and preventing interference with other communications. Li-Fi also helps conserve the radiofrequency spectrum, addressing challenges Wi-Fi faces in increasingly crowded environments.

Inside buildings without wired connections, PLC infrastructure can enable Li-Fi Access Point connections to the LAN through the electrical lighting system.

The primary focus of Li-Fi technology is to eliminate radio-wave emissions.

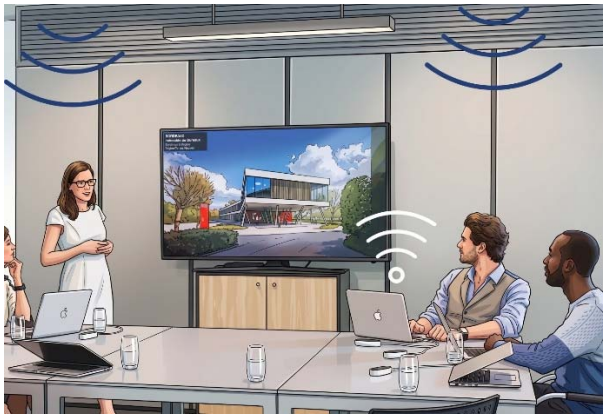


Figure 2: Li-Fi illustration

4.2.2.2 Standardization

The main standards for Li-Fi technology are Recommendation ITU G.9991 [i.3] and IEEE 802.11bb [i.2].

4.2.2.3 Constraints Related to Buildings' Topology and Uses

The elementary limitation of Li-Fi technology is the requirement for a USB key with an LED emitter/transmitter connected to a PC or tablet. The issue is that young students can easily unplug these devices, which may be lost.

4.2.2.4 General Recommendations

When manufacturers of tablets or PCs integrate a Li-Fi sensor into their devices, it will help facilitate the adoption of this technology in educational institutions.

4.2.2.5 Specific Needs of French Elementary Schools

Despite the limitations of using an external Li-Fi USB dongle, Li-Fi remains an excellent way to fully restrict young students' exposure to radio waves, in accordance with young children's health in general and with the French Abeille Law [i.13].

4.3 Wired buildings

4.3.1 RJ45 cabling + Wi-Fi® Hotspot

4.3.1.1 Fundamental Principles

A Power over Ethernet (PoE) switch in the telecom cabinet supplies power and connectivity to Wi-Fi hotspots, ensuring coverage across all classrooms. Two or three classrooms can share a single wireless access point.

4.3.1.2 Standardization

Wi-Fi standards are established by IEEE under the 802.11 protocols. These standards define the protocols and procedures that wireless devices use to communicate and connect to the internet. The evolution of these standards has led to significant improvements in speed, range, and reliability.

4.3.1.3 Constraints Related to Buildings' Topology and Uses

Building wiring should adhere to specific guidelines to ensure optimal performance and accommodate future uses that require increased bandwidth. The increasing demand for power for new equipment using the PoE IEEE 802.3af [i.7], (clause 33), IEEE 802.3at [i.8] and IEEE 802.3bt [i.9] standards should be considered when selecting the appropriate cable and PoE switches with a compliant PoE budget.

4.3.1.4 General Recommendations

Unlike corporate, college, or university buildings, elementary school buildings should prioritize basic equipment.

When a Wi-Fi Hotspot is installed in each classroom, its Wi-Fi signal should be set to the minimum strength.

Wi-Fi terminals should not be installed at head level for people, especially children. Instead, they should be installed higher up, such as at the ceiling or inside false ceilings.

Another option is to place the Wi-Fi Hotspot in the corridor, equidistant from 2 or 3 classrooms, while respecting the previous point about the installation height.

Last but not least, Wi-Fi should be disabled on all tablets at the end of a connected pedagogical activity, preferably automatically.

4.3.1.5 Specific Needs of French Elementary Schools

As previously indicated, Wi-Fi Hotspots are installed in the ceiling or at ceiling level, making them difficult to reach and turn off.

Specific devices or software tools should be installed to enable users to easily control Wi-Fi emissions during Wi-Fi-enabled digital activities. On the other hand, it is highly recommended that editors include a command in their pedagogical application to automatically enable and disable the tablets' Wi-Fi. Otherwise, there is no point in disabling just the Wi-Fi access point when 20 or 30 tablets are still transmitting.

4.3.2 RJ45 cabling + Li-Fi

4.3.2.1 Fundamental Principles

There is no difference between a PLC and a wired infrastructure; however, a wired infrastructure offers better Li-Fi performance due to its higher shared bandwidth.

4.3.2.2 Specific Needs of French Elementary Schools

As noted in clause 4.2.2.5, Li-Fi is the most effective means of completely suppressing electromagnetic emissions. It is a complementary solution to Wi-Fi. Both technologies should be used in schools: Wi-Fi in elementary schools and Li-Fi in nurseries for younger children.

However, compared with Wi-Fi, Li-Fi is more costly to deploy because it requires dongles and Access Points to connect Li-Fi transceivers to the LAN.

For widespread adoption in schools, Li-Fi technology requires PC and tablet manufacturers to integrate Li-Fi receivers into their devices.

5 Specific configurations to be used in French Schools

5.1 Official constraints

5.1.1 Abeille Law requirements (extract)

Law No. 2015-136 of 9 February 2015 relating to sobriety, transparency, information, and consultation in matters of exposure to electromagnetic waves [i.13].

Article 7:

"I. - "…" The installation of fixed terminal equipment with wireless internet access is prohibited in areas designated for the reception, rest, and activities of children under three years old.

II. - In primary school classrooms, wireless access to the equipment "..." are deactivated when not in use for digital educational activities.

III. - In primary schools, any new radio network installation is to be notified to the school council beforehand."

Article 4:

""…"

3° Establishments offering Wi-Fi access to the public clearly display this with a pictogram at the entrance."

5.1.2 Ministry of Education

The French Ministry of Education has released a digital reference guide outlining best practices [i.14] for implementing Wi-Fi® infrastructure. The guide also addresses legal [i.12] and health [i.11] considerations for Wi-Fi use, particularly in primary schools (digital reference guide outlining best practices [i.14] within three references are defined – technic, health, legal, see descriptions done in annexes A, B, and C).

5.2 METHOD #1 – Micro Smart Plug

5.2.1 Fundamental Principles

Compatible with certain smart home and building protocols, such as EnOcean [i.1], some Micro Smart Plugs allow to turn any connected device on or off. It may be tempting to connect the AC/DC power supply directly to a Wi-Fi device.

5.2.2 General Recommendations

Such a solution is not recommended:

- This requires running costly high-current cables to the power outlets of each Wi-Fi Hotspot.
- Without a microcontroller, these devices cannot support advanced functions. Specifically, nothing prevents the Wi-Fi hotspots from being accidentally turned off, potentially rendering them out of service.

5.3 Method #2 - Centralized Wi-Fi® management using a controller

5.3.1 Wi-Fi Hotspot managed with Control command

5.3.1.1 Fundamental Principles

A central controller is connected to the LAN inside the telecom cabinet. It is linked to either physical RJ45-wired buttons or virtual buttons in each classroom or on the teacher's laptop screen. When the teacher presses these buttons, an IP message is sent to the controller indicating the classroom where the signal originates and which Wi-Fi Hotspot needs its Wi-Fi turned on or off.

With this solution, the controller enables IT managers to access a wealth of data to analyse the use of the wireless infrastructure.

5.3.1.2 Standardization

No standard defines the Wi-Fi control commands, which can vary from one manufacturer to another.

5.3.1.3 Constraints Related to Buildings' Topology and Uses

If the customer needs to use physical buttons, each should be wired using an RJ45 cable.

5.3.1.4 General Recommendations

Such a solution is more advantageous in colleges and universities where Wi-Fi needs to be enabled most of the time.

5.3.1.5 Specific Needs of French Elementary Schools

Under the French Abeille Law [i.13], teachers need to press physical buttons to easily turn on Wi-Fi. Therefore, this solution requires increasing the number of RJ45 wires.

On the other hand, the fact that control commands can be modified after implementing a new Wi-Fi Hotspot or a new release necessitates additional maintenance. Consequently, this kind of solution often requires an annual licensing agreement to ensure the best ongoing service in schools.

5.4 Method #3 – Decentralized Wi-Fi® management using unconnected buttons

5.4.1 Decentralized Wi-Fi® management

5.4.1.1 Fundamental Principles

Unlike Method #2 and #1, which use RJ45 cables, small "controllers" are plugged between the LAN wall outlet and the Wi-Fi Hotspot. These controllers are paired with a button via a specific protocol, such as EnOcean [i.1], a radio-based home automation protocol. Controllers should receive signals from the buttons up to 30 meters away to turn on the Wi-Fi Hotspot by switching off the PoE supply.

These buttons should preferably use a wireless energy-harvesting protocol to eliminate the need for wires or batteries. Like the EnOcean [i.1] protocol, the chosen solution should be as unobtrusive as possible to minimize its electromagnetic emissions. Therefore, any "mesh" solution should be avoided.

5.4.1.2 Standardization

The EnOcean [i.1] protocol is defined by ISO/IEC 14543-3-10 [i.4], making it the first wireless energy-harvesting protocol to achieve ISO international standard compliance.

5.4.1.3 Constraints Related to Buildings' Topology and Uses

Depending on the topology and wall thickness, using a Wi-Fi hotspot in each classroom is preferable.

If one Wi-Fi hotspot can cover 2 or 3 classrooms, it would significantly reduce installation costs by decreasing the number of equipment units, cabling, and RJ45 plugs.

5.4.1.4 General Recommendations

- Do not use batteries to reduce human interventions, costs, and environmental impact.
- Conduct a brief coverage study to assess the feasibility of sharing Wi-Fi hotspots among multiple classrooms.
- Place buttons near light switches in accordance with standards relating to people with disabilities.
- Consider using wireless communication via the EnOcean [i.1] protocol between the technician's PC and the mini controllers to simplify installations. In this case, include an EnOcean [i.1] antenna on the technician's PC.
- Since Wi-Fi hotspots are disabled by default, a simple command should be available to enable all hotspots at once, simplifying administration.
- A function should be implemented to prevent premature shutdown of Wi-Fi terminals during initialization.
- The ability to define operating time slots is essential. Likewise, the system should offer time delays or automatically shut down Wi-Fi in case of forgetfulness.
- Finally, a dedicated IP/EnOcean [i.1] gateway is highly recommended to enable remote management of all classrooms and schools from a central IT site. Furthermore, such a solution could also enable the collection of environmental data from various sensors (temperature, humidity, energy consumption, radon, or CO_2).

6 Conclusion

The purpose of the present document is to support the marketing of solutions that follow recommendations, meet user expectations, and promote best practices.

In the education sector, specifically primary and nursery schools, methods #2 and #3 could benefit from a shared foundation for manufacturers and software developers.

For example, under method #2, to remotely stop electromagnetic emissions from Wi-Fi® routers without turning them off, control commands vary by manufacturer. It would be useful to add a standardized control command common to all manufacturers.

The granularity of wireless infrastructure should be considered when collecting environmental data from all parts of buildings. This data should be valuable not only for technical services that monitor air quality and other factors but also for teachers who could use it as new pedagogical material to illustrate lessons on ecological transition.

Methods #2 and #3 should identify the most effective way to integrate into an IoT ecosystem.

Annex A:
Guidelines for the use of Wi-Fi® in establishments and schools - Technical framework

This annex reviewed the main sections and key points from the technical and security framework for Wi-Fi® usage in schools [i.10]. The summary below captures the document's structure, main recommendations, and actionable insights, focusing on the technical, organizational, and security aspects relevant to educational institutions.

It provides an actionable overview for IT managers, school administrators, and technical staff responsible for secure Wi-Fi deployment in educational settings.

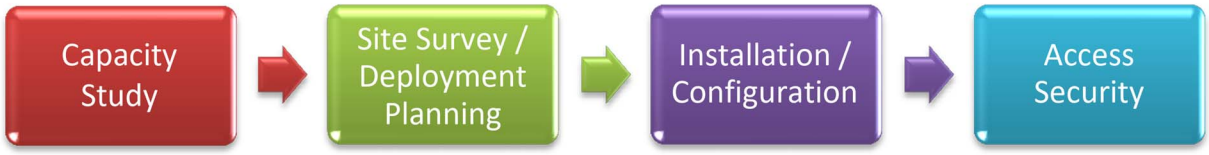


Figure A.1

It covers the following key areas:

1) **Technical Framework and Capacity Study:**
Before implementing Wi-Fi, it is essential to assess the available and required bandwidth, considering both wired and wireless technologies. The study should determine the number of users, their distribution, and the types of applications used to ensure sufficient quality of service.

2) **Site Survey and Deployment Planning:**
A site survey is necessary to evaluate the feasibility of Wi-Fi deployment, taking into account building architecture, potential obstacles, and legal constraints. The goal is to determine the optimal placement and number of access points, ensuring adequate coverage and compliance with regulations.

3) **Installation and Configuration Best Practices:**
The document details Wi-Fi standards and emphasizes the importance of using up-to-date, secure technologies. Best practices include enabling Wi-Fi only when needed, disabling automatic connections, keeping systems updated, and avoiding connections to untrusted networks.

4) **Access Security:**
Strong authentication and encryption are critical. WPA2 with AES-CCMP encryption is strongly recommended. For professional environments, centralized authentication (WPA-Enterprise) should be used. Passwords are recommended to be robust and to be changed regularly. The use of WEP and WPA (without WPA2) is discouraged due to security vulnerabilities.

5) **Risk Assessment and Security Recommendations:**
The document identifies key risks, such as data leakage, unauthorized network use, exposure of minors to inappropriate content, and operational disruptions. It recommends a risk-based approach, applying 23 specific security recommendations from the French national cybersecurity agency (ANSSI), including network segmentation, regular updates, and strict access controls.

6) **Network Architecture and Segmentation:**
Wi-Fi networks should be segmented from wired networks, with strict filtering and monitoring. Guest networks should be isolated and have more restrictive security policies. Captive portals can be used for visitor access but have inherent security limitations.

7) **Glossary and References:**
The document includes a glossary of technical terms and references to relevant legal and technical guidelines.

Example Implementation:

When deploying Wi-Fi in a school, conduct a bandwidth assessment, perform a site survey, use WPA2-Enterprise with strong authentication, segment the Wi-Fi from the main network, and regularly update all equipment to ensure both technical and organizational security.

Annex B:
Guidelines for the use of Wi-Fi® in establishments and schools - Radio frequencies and health

This annex reviews the main sections and key points, focusing on the scientific, legislative, and practical recommendations regarding Wi-Fi® use in schools, particularly with respect to health and legal requirements [i.11].

The summary below captures the essential findings and guidance provided by the document covering the following key areas:

1) **Scientific Expertise on Radiofrequencies and Health:**

   - The French Agency for Food, Environmental and Occupational Health & Safety (ANSES) reviewed over 1 000 studies and found no sufficient evidence of non-carcinogenic health effects from radiofrequencies, including Wi-Fi, on the central nervous system or other organs.
   - Some studies suggested a possible carcinogenic risk from intensive, long-term use of mobile phones held to the ear, but these findings do not apply to Wi-Fi exposure.
   - Wi-Fi emits, on average, ten times less radiation at maximum power than a mobile phone, and exposure decreases rapidly with distance. Wi-Fi access points do not emit continuously, and exposure is minimal when not in use.
   - ANSES does not recommend any specific precautions or restrictions regarding Wi-Fi use, regardless of age, including for children.

2) **Considerations for Children:**

   - There is no evidence identifying children as more sensitive to radiofrequencies, but ANSES recommends further research.
   - The only recommendation for children is to encourage moderate mobile phone use and prefer hands-free devices with low Specific Absorption Rates (SAR). This does not concern Wi-Fi.

3) **Legislative Aspects:**

   - French law prohibits mobile phone use by students during teaching activities in schools and colleges, which indirectly limits Wi-Fi access via smartphones during class.
   - Wi-Fi access points are prohibited in areas for children under three years old (such as nurseries), but not in primary schools.
   - In primary school classrooms, Wi-Fi equipment are deactivated when not used for educational digital activities, and school councils are to be informed before installing wireless networks.

4) **Recommendations:**

   - Turn off Wi-Fi access points in primary school classrooms when not needed for teaching.
   - Deactivate Wi-Fi on mobile devices when not required for educational activities.
   - Follow technical guidelines for installing and configuring Wi-Fi to optimize security and minimize exposure.
   - Conduct site studies to optimally place Wi-Fi access points.

Key Takeaway: The document concludes that, based on current scientific evidence and legislation, Wi-Fi can be used in schools with simple precautionary measures, mainly focused on minimizing unnecessary exposure and ensuring proper technical setup, but there is no evidence requiring strict restrictions or bans.

Annex C:
Guidelines for the use of Wi-Fi® in establishments and schools - Uses and legal framework

This annex provides a comprehensive framework for the deployment and use of Wi-Fi® in schools, focusing on legal, technical, and organizational aspects [i.12]. Its main goal is to help school leaders, IT directors, and local authorities implement reliable, secure Wi-Fi infrastructure tailored to the needs of educational institutions.

- **Purpose:** The document aims to guide the implementation of Wi-Fi in schools, ensuring that all legal, pedagogical, and technical requirements are met.
- **Audience:** It is primarily intended for school principals, IT directors, and local authorities responsible for digital infrastructure in schools.
- **Essentials:**
  - Wi-Fi can be installed in schools and educational institutions.
  - Needs assessment should address who needs to do what, where, and how.
  - Installation should comply with technical and security recommendations.
  - Ethical and regulatory aspects specific to Wi-Fi should be included in the institution's usage charter.
  - General terms of use are to be established and communicated to users, and acceptance should be mandatory before network access.

Framework Organization:



Figure C.1

- The document is structured around four main contexts:
  1) **Legal Context:** Covers laws and obligations related to Wi-Fi deployment, including health and safety regulations and data protection.
  2) **Usage Context:** Defines user groups (students, staff, visitors, BYOD), their needs, and the types of devices and services accessed.
  3) **Regulatory Context:** Addresses compliance with national laws, such as restrictions on Wi-Fi in certain areas and requirements for user information.
  4) **Technical Context:** Details technical studies, equipment placement, configuration, and security measures for safe and effective Wi-Fi deployment.

Implementation Guidance:

- The document provides step-by-step guidance, from assessing needs and defining use cases to technical deployment and securing access.
- It emphasizes the importance of involving all stakeholders in the planning process and adapting solutions to local conditions.
- Special attention is given to the protection of personal data, prevention of unauthorized Wi-Fi access points, and the inclusion of specific clauses in the institution's digital usage charter.

Conclusion: This reference serves as a national framework to support the responsible and effective implementation of Wi-Fi in schools, ensuring both compliance with legal requirements and the fulfilment of educational needs.

History

Version	Date	Status
V1.1.1	September 2026	Publication